

Soft Robotics and Neurodegenerative Disease: Enhancing Sleep and Mental Wellbeing in Dementia Patients through Innovative Care Solutions

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Figure 1: A sample of the study process and artefacts

Abstract

Soft robotics offers significant promise for enhancing healthcare, providing benefits such as improved rehabilitation and daily assistance for both physical and mental health. Despite these advantages, its application in caring for older adults with neurodegenerative disorders (NDD) remains largely unexplored, even though the World Health Organization (WHO) identifies NDD as a major global health burden. This study investigates the perceptions of soft robotics among individuals with neurodegenerative diseases. Through focus groups, individual interviews, and co-design sessions with patients, clinicians, and caregivers, we explored usability, safety, concerns, emotional comfort, and design ideas for soft robotics. Our findings highlight the necessity for user-centered design to address these concerns, emphasizing the importance of creating multi-purpose technologies that can be used by both patients and

caregivers to support each other and build relationships. This approach aims to enhance the acceptance and effectiveness of soft robotic technologies for this vulnerable population.

CCS Concepts

• Co-design; • Soft robotics; • Sleep and wellbeing; • Neurodegenerative Disease;

Keywords

Soft robotics, neurodegenerative diseases, co-design, participatory design, design methods

ACM Reference Format:

Bijetri Biswas, Elizabeth Coulhard, and Anne Roudaut. 2024. Soft Robotics and Neurodegenerative Disease: Enhancing Sleep and Mental Wellbeing in Dementia Patients through Innovative Care Solutions. In *Companion of the 2024 Computer-Supported Cooperative Work and Social Computing (CSCW Companion '24)*, November 9–13, 2024, San Jose, Costa Rica. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3678884.3685937>



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CSCW Companion '24, November 9–13, 2024, San Jose, Costa Rica
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ACM ISBN 979-8-4007-1114-5/24/11
<https://doi.org/10.1145/3678884.3685937>

1 Introduction

In recent years, neurodegenerative diseases, including dementia, have become more prevalent with the increasing ageing population. The National Health Service (NHS) faces pressing demands

in the UK, and providing appropriate care for patients over 65 is increasingly challenging. The rising demand for caregivers makes it impractical to offer continuous assistance. Hence, many people with dementia live independently or in residential homes. The isolation experienced by many patients due to their condition significantly affects their mental health [12].

Soft robotics is an emerging field in healthcare technology that promises significant advancements in patient care. Unlike traditional rigid robots, soft robots are designed to interact more naturally and safely with the human body. The unique characteristics of soft robotics make it particularly suited for delicate tasks and, importantly, for serving vulnerable populations. The flexibility and adaptability of soft robotics offer unique benefits for enhancing both physical and mental health, a critical aspect of comprehensive patient care [6].

Despite these potential advantages, the application of soft robotics in the care of older adults with neurodegenerative diseases (NDD) remains underexplored. Neurodegenerative disorders are among the leading causes of death and disability worldwide. The World Health Organization (WHO) highlights these conditions as a major global health burden, stressing the urgent need for innovative solutions to improve patient quality of life and care [1]. Current caregiving robots, typically programmed to meet patient needs, are expected to become more prevalent in the coming years. However, their rigid structure limits safe human-robot interaction [2]. In contrast, soft robotics, which involves constructing and designing robots from flexible materials and electronics, promises safer interactions by optimising control and adaptability to human needs [14].

There's a significant gap in current research in understanding how soft robotics is perceived by those directly affected by neurodegenerative disorders—patients, caregivers, and healthcare professionals. Their insights are crucial for developing technologies that are not only effective but also acceptable and user-friendly. Understanding these perceptions can guide the design of soft robotic systems to meet the needs and preferences of end-users better. Therefore, this study aims to bridge this gap by investigating the perceptions and requirements of individuals with neurodegenerative diseases regarding soft robotics. We conducted focus groups, individual interviews, and co-design sessions with patients, clinicians, and caregivers.

Our research delves into multiple critical aspects, including usability, safety, emotional comfort, social factors and innovative design concepts, to gather comprehensive insights that will inform the development of soft robotics. By examining these factors, we shared design ideas to create soft robotic solutions that meet the practical needs of users and ensure their safety and emotional well-being. This holistic approach is intended to enhance care practices, making soft robotics a more effective and accepted tool for supporting individuals with neurodegenerative disorders. Through this detailed exploration, we seek to foster a better understanding of how these technologies can be seamlessly integrated into everyday care, ultimately improving the quality of life for both patients and caregivers.

2 Related Work

Research studies have shown that soft robotics can be used for personal care in the elderly population. A study explores the development and application of soft robotics to assist elderly individuals with the task of undressing. The researchers designed a soft robotic system capable of gently and efficiently helping users remove clothing without causing discomfort or requiring significant physical effort [4]. The system effectively promoted independence and reduced reliance on caregivers for daily activities such as dressing and undressing.

Another paper discusses another assistive soft robot aimed at providing personal care for elderly individuals. The robot incorporates soft materials to ensure safe and comfortable interaction with users [7]. The study highlights the robot's ability to perform various personal care tasks, such as feeding and mobility assistance, demonstrating its potential to enhance the quality of life for older adults.

Other types of soft robotics include arm-assistive medical devices used to aid individuals with limited arm mobility. The device, designed to be lightweight and user-friendly, assists users in performing arm movements necessary for daily activities. The study found that the soft robotic device significantly improved users' ability to perform tasks independently, enhancing their autonomy and reducing the need for caregivers [9].

In recent years, a wearable elbow exoskeleton has been created using soft robotics technology. It involves employing shape memory alloy wire actuators [13]. The exoskeleton is designed to assist with elbow movements, providing support and enhancing the user's range of motion. The study concluded that the exoskeleton effectively augmented the user's elbow function, showing promise for rehabilitation applications and daily assistance for individuals with impaired arm mobility.

Soft robots designed to aid sleep and well-being, such as the Somnox Sleep Robot, are gaining attention for their ability to simulate breathing patterns and produce soothing sounds to help users relax and sleep better. These robots are typically made from soft, flexible materials to provide comfort and a sense of calm, mimicking the rise and fall of human breathing. Somnox, a Dutch startup, is a notable example of this technology, focusing on improving sleep quality as an alternative to pharmaceutical intervention [11].

A study investigated the effects of a 3-week intervention using the Somnox sleep robot in adults with insomnia. It was single-case design with four participants. The study measured sleep quality, arousal, anxiety, and depression through sleep diaries, actigraphy, and questionnaires. The results showed some participants had improvements in sleep and reduced arousal. The intervention was deemed safe, with no adverse effects on mental health [11]. However, the mixed results suggest that larger studies are needed to validate the effectiveness of the sleep robot. Overall, while soft robots like Somnox have shown promise in enhancing sleep and overall well-being, there is a clear need for innovation to extend these benefits to neurodegenerative patients.

Despite the advancements in sleep and well-being robots, a notable gap exists in developing soft robotics targeting patients with neurodegenerative diseases. Most current soft robotic applications for healthcare are not designed to address the unique needs of this

population. These patients often require devices that can assist with their specific symptoms, such as tremors, rigidity, or cognitive impairments, which are not currently the focus of existing sleep robots. This could involve adapting the technology to support better the physical and neurological challenges these individuals face, ensuring they can also benefit from the advancements in soft robotics.

2.1 Method

In this preliminary study, we aimed to enhance our comprehension by engaging in semi-structured interviews with a diverse range of stakeholders, including clinicians, patients, and caregivers. We aimed to gather insights into user needs, preferences, and potential design concepts. To mitigate any potential unfamiliarity with soft robotics technology, we ensured participants were briefed on the topic before the interviews commenced. Furthermore, this project received ethical clearance from the University of Bristol’s ethics review board.

2.2 Participants

We recruited 5 older adults diagnosed with neurodegenerative diseases such as mild cognitive impairment, dementia, Parkinson’s disease, and Alzheimer’s disease, ensuring that all participants were capable of giving informed consent. Additionally, we included 5 caregivers who identified themselves as such, partners, and professionals responsible for caring for these individuals. 5 Clinicians, including clinical researchers, occupational therapists, physiotherapists, neurologists, and nurses, were also part of the study. Participants were specifically selected for their limited or no prior knowledge of soft robotics to ensure we captured a wide range of unbiased perspectives. All participants were recruited through charities, personal contact or from Southmead Hospital, Bristol. Recruitment focused on ensuring a diverse representation of individuals from different ethnic backgrounds, experiences and perspectives to enrich the data collection.

2.3 Research Process

2.3.1 Phase 1: Semi-structured interview (20-30 minutes): At first, the researcher welcomed participants, explained the purpose and context of the study, and facilitated introductions. Participants had the opportunity to ask questions or express any concerns before signing the consent form. Each participant received a copy of the consent form, with another copy securely stored on university premises. Participants were informed that audio recording would be used for data collection, and their consent was obtained. This session aimed to gauge participants’ prior knowledge and perceptions of soft robotics and the potential applications they envisioned.

2.3.2 Phase 2: Technology demonstrations (40-60 minutes). Participants were shown videos and pictures of examples of soft robotics. They were then asked to freely discuss their opinions about the technology and express any concerns they might have. They also had the opportunity to share any experiences with similar or other technologies. Lastly, the general perspective of soft robotics technology, what are the benefits and concerns.

2.3.3 Phase 3: Ideation (60 minutes). Participants engaged in an ideation session where they were asked to sketch or describe ideas on how soft robotics could assist people with neurodegenerative diseases. They were encouraged to explain how these ideas could help address specific issues related to their conditions. Participants also had the opportunity to draw some of their design ideas.

Due to the nature of the participants, all parts of the research were conducted remotely, subject to availability and convenience.

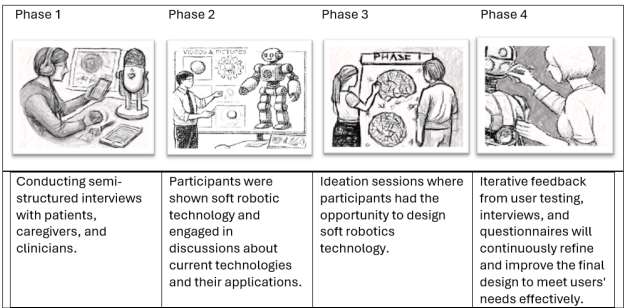


Figure 2: Research Process

2.4 Data analysis

We analysed our data from Phases 1 and 2 using affinity diagramming, a bottom-up approach to organising qualitative data. Phase 3, the co-design session, is recorded on the Miro board, and images are hand-drawn and sent. We conducted the interviews and technology demonstration via Zoom or Microsoft Teams. Due to the complex needs of their conditions, we assisted individuals remotely as much as possible regarding the drawing.

3 Preliminary results

3.1 Phase 1: Interview summary of thematic analysis

The findings from phase 1 highlight a significant knowledge gap regarding soft robotics, with 90% of the participants reporting limited familiarity with the topic. Participants expressed curiosity and interest in learning more about soft robotics, often citing the intriguing nature of the term itself. When asked to define or describe soft robotics, responses varied widely. A person with the condition reported, "No clue, like a pillow or something?". This reflects a general lack of awareness and understanding of the technology among patients, consistent with previous studies that found low awareness of emerging healthcare technologies in older adults [5].

While another clinician likened it to existing robotic technologies, stating, "Something like Paro robot", referring to the well-known therapeutic robot used for dementia care [8]. Another clinician mentioned, "Recently, one of my patient’s family bought robotic gloves and told us to see whether they could be useful for upper limb exercises," indicating a more hands-on familiarity and openness to integrating new technologies into patient care.

Clinicians also associated soft robotics with safety and potential utility for patient care, with one stating, "It reminds me of prosthetics," emphasising the potential for these technologies to provide

assistive functions. Another clinician noted, "It sounds safe and useful potentially for patients," underscoring the perceived safety and applicability of soft robotics in clinical settings. This aligns with findings from [3], who reported positive clinician attitudes towards assistive technologies that promise enhanced safety and functionality. Caregivers mentioned conducting pre-interview research "I searched read about soft robotics online" to familiarise themselves with the concept. This indicates a proactive approach to engaging with unfamiliar topics and technologies. This is due to a willingness to benefit their loved ones, a sentiment echoed in studies on caregiver adoption of assistive technologies [10]. These insights highlight the initial perceptions and expectations surrounding soft robotics among participants, providing valuable context for the subsequent phases of the study.

3.2 Phase 2: Summary from the technology demonstration

3.2.1 Understanding through Visuals: Participants found videos helpful in comprehending the concept of soft robotics, indicating that visual aids effectively elucidated complex ideas.

3.2.2 Concerns and Experiences with Assistive Technology: Individuals with conditions expressed apprehensions and shared personal experiences with robotic assistive technology. One individual expressed fear and frustration due to previous malfunctions, stating, "they find it scary to use it because it broke few times and it was not listening to them". One said, "how are this technology robotics they don't look like robots" Another participant associated robotic technology with frustration, comparing it to Alexa, noting, "it does not listen and build frustration." There were concerns about cost and durability, with one participant expressing scepticism, "It will cost a lot, and they will probably break". Additionally, participants highlighted the need for technology to aid memory and prompt daily tasks, with one stating, "I need something that will prompt me to do things, help me sleep and cheer me and my wife."

3.2.3 Caregiver Perspectives: Caregivers emphasised the lack of attention given to their needs in technology design. They expressed concerns about the difficulty of using and repairing devices, particularly considering their own age. One caregiver mentioned, "when they break we don't know where to go and fix as I am quite old as well." They emphasised the importance of devices that assist with sleep and daily tasks, with specific reference to the weight and cost of existing options. Another caregiver mentioned, "Anything that could help both of us to sleep and do chores like read books and use with the instruction from Alexa would be great."

3.2.4 Clinician Insights: Clinicians recognised the potential of soft robotics in addressing various symptoms and enhancing the quality of life for patients. They emphasised the importance of simplicity, versatility, and user-friendliness in device design. One clinician stated, "if this kind of technology can help people to sleep and mood would be useful," while another highlighted the need for devices to assist with communication and mood regulation, saying, "calm them during the time when they find it difficult to control their mood, change in personality". Another clinician said "anything that can detect their environment and identify trigger point, like why they not sleeping or change in weight". One suggested, "something

to help with speech when they lose track of conversation like hint with words or assist with language". Also, they "need something that both can use the careers and people to help them with their stress, anxiety and wellbeing". Overall, the quotes highlight the diverse perspectives and needs of participants, underscoring the importance of addressing user concerns in the development of soft robotic technologies.

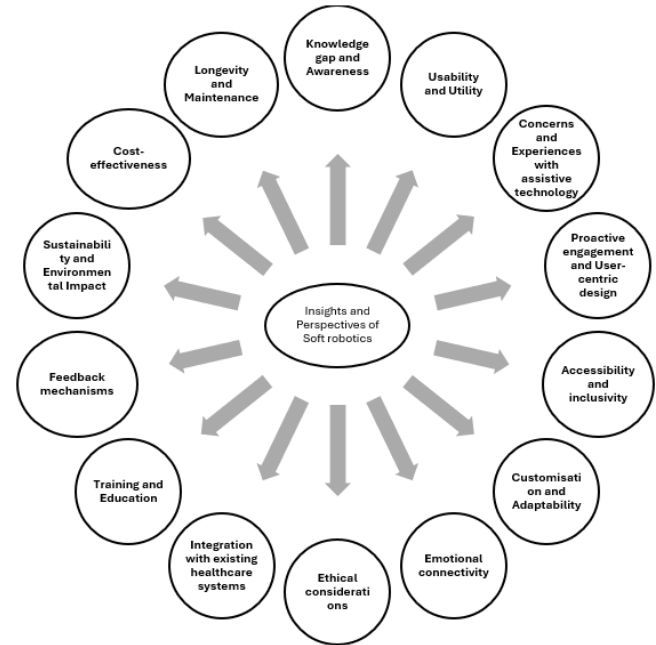


Figure 3: Combined thematic diagram of phase 1 and 2

3.3 Phase 3: Ideation

Based on the suggestions and concerns raised by participants, here are two design ideas for soft robotics shown in figure 4.

3.3.1 Functionalities and Technical Aspects of Soft Robotic Solutions. The left image in figure 4 is a humanoid robot named ComfortBot, which features a rounded, cushioned exterior designed to maximise safety and comfort, making it highly user-friendly. Its friendly digital face with expressive eyes and a gentle smile enhances emotional engagement, encouraging users to interact with the robot. The padded body, with segmented arms and legs, allows for flexible and smooth movements, mimicking human-like gestures. Integrated sensors monitor environmental factors such as temperature, humidity, and air quality, ensuring a safe and comfortable environment. Ambient LED lights enhance ComfortBot's aesthetic appeal and signal different interaction modes, such as sleep or active engagement. The robot's actuation and control systems use soft actuators for gentle movements, powered by a rechargeable battery optimised for extended use. Connectivity with smart home systems further integrates the robot into the user's daily life, providing automated responses to environmental changes.

The right below shows a robot named MoodBot, which is designed with a cushioned exterior and small retractable legs for

smooth locomotion, making it both compact and friendly in appearance. It features vents for diffusing pleasant smells, using integrated scent diffusion technology to promote relaxation and a pleasant environment for dementia patients. MoodBot's simple, friendly design, coupled with a digital face displaying calming expressions, helps soothe and comfort users, particularly those with dementia. It includes memory aid functions, such as playing familiar sounds or music to stimulate memory and provide reassurance. The mobility mechanism combines soft robotics with traditional methods for smooth movement across various surfaces. Controlled via a smartphone app, caregivers can customise settings and monitor the robot remotely. Durable, washable materials ensure easy maintenance and longevity, making MoodBot an ideal tool for improving mood and sleep in dementia care settings.

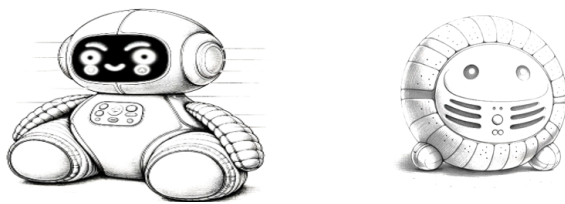


Figure 4: Initial design ideas

3.4 Phase 4: Iterative Feedback and Refinement

In Phase 4, we plan to implement an iterative feedback loop to refine the final designs of ComfortBot and MoodBot. This phase will involve feedback sessions where caregivers, dementia patients, and clinicians will interact with the robot in real-world scenarios. Feedback will be collected through demonstration, structured interviews focusing on usability, comfort, emotional engagement, and overall effectiveness. We will utilise this feedback to make improvements, addressing any identified issues and enhancing the robots' functionalities. This process will help us achieve a design that is not only technically sound but also deeply resonant with its intended users.

4 Discussion

The thematic analysis reveals key insights into participants' perceptions of soft robotics. Phase One highlighted a significant knowledge gap, with 90% of participants unfamiliar with the topic yet curious to learn more. Clinicians saw potential safety and utility for patient care, while caregivers proactively researched the topic. Phase Two findings emphasised the effectiveness of visual aids, concerns about the reliability and cost of assistive technology, and the need for devices to aid memory and daily tasks. Caregivers and clinicians stressed the importance of user-friendly designs that enhance patient quality of life and creating technology targetted at carers. Also, suggesting features for communication, mood regulation, and environmental monitoring for individuals with the condition. These insights underscore the necessity of addressing user concerns in developing soft robotic technologies not only for patients but also for caregivers.

Most previous research on neurodegenerative diseases has yet to explore the application of soft robotics and its potential benefits. Traditional studies focus primarily on pharmaceutical treatments, cognitive therapies, and diagnostic techniques. While soft robotics is an emerging field, it has inherent challenges, such as the necessity of incorporating hard components, which can limit their flexibility and adaptability. Additionally, recent designs of soft robotic devices, such as those aimed at improving sleep quality, have yet to be tailored to meet the specific needs of individuals with neurodegenerative diseases. This gap highlights the need for innovative approaches to integrate soft robotics into therapeutic and supportive interventions for this population.

The emerging field of soft robotics holds promise in addressing critical challenges faced by individuals, and researchers are evolving alongside this technology. However, as soft robotics is still in its infancy, there is a pressing need for additional research to comprehensively explore its potential, especially in addressing the needs of clinical and community-dwelling NDD populations. Our study represents a pioneering effort in this regard, aiming to bridge the gap between technology design and user requirements in a diverse population. By adopting a transdisciplinary approach and collaborating across different disciplines, we seek to develop informed solutions to the current challenges in health technology, particularly concerning redundancy and cost-effectiveness. Our project aims to create low-cost, sustainable soft robotic technologies that can be rapidly personalized to meet the unique needs of users. Additionally, our research will contribute to a deeper understanding of human-soft-robot interaction and the associated challenges, further enhancing the development and deployment of these technologies in clinical and community settings.

Our preliminary findings offer valuable insights into the perceptions and potential applications of soft robotics in caring for individuals with neurodegenerative diseases. It is essential to acknowledge the limitations of this initial study. The relatively small sample size and the qualitative nature of the data require further research to confirm and expand upon our results. It is an ongoing study, and we are still recruiting individuals. Additionally, recruiting patients with neurodegenerative conditions presents significant challenges due to their cognitive vulnerabilities and potential for rapid deterioration. Ensuring informed consent at every study stage is crucial to confirming participants' capacity and understanding of the study's expectations. More comprehensive studies involving diverse populations are essential to validate the findings and refine the proposed designs. In future, we plan to incorporate iterative design cycles to enhance the robustness and applicability of soft robotic solutions. Addressing these limitations can provide a more detailed understanding of the technology's effectiveness and acceptance, ultimately leading to more effective and user-friendly innovations in the care of neurodegenerative patients.

5 Conclusion

Our findings underscore the importance of user-centred design in developing soft robotics for individuals with neurodegenerative diseases. By addressing concerns and incorporating multi-purpose functionalities, we aim to create technologies that support patients and facilitate collaborative use with caregivers, fostering

relationships and improving overall care experiences. This paper contributes to the CSCW community by providing valuable insights into designing soft robotics for vulnerable populations, ultimately aiming to enhance the acceptance and effectiveness of these technologies.

6 Acknowledgments

We thank the patients and carers at Parkinson's UK and Southmead Hospital ReMemBr Group for their time and advice on this project. This work was supported EPSRC Digital Health and Care Centre for Doctoral Training (CDT) at the University of Bristol (UKRI Grant No. EP/S023704/1) as well as EP/P004342/1. Opinions, findings, and conclusions or recommendations expressed in this paper are those of the authors and do not necessarily reflect any organisational views.

References

- [1] Suman Dutta, Miriam Sklerov, Charlotte E Teunissen, and Gal Bitan. 2023. Trends in biomarkers for neurodegenerative diseases: Current research and future perspectives. *Frontiers in aging neuroscience* 15 (2023), 1153932.
- [2] Justyna Gerłowska, Marzena Furtak-Niczyporuk, and Konrad Rejdak. 2020. Robotic assistance for people with dementia: a viable option for the future? *Expert Review of Medical Devices* 17, 6 (2020), 507–518.
- [3] Christopher M Hayre, Dave Muller, Marcia Scherer, Paul MW Hackett, and Ava Gordley-Smith. 2024. Emerging Technologies in Healthcare: Interpersonal and Client Based Perspectives. (2024).
- [4] Tim Helps, Majid Taghavi, Sarah Manns, Ailie J Turton, and Jonathan Rossiter. 2018. Easy undressing with soft robotics. In *Towards Autonomous Robotic Systems: 19th Annual Conference, TAROS 2018, Bristol, UK July 25-27, 2018, Proceedings 19*. Springer, 79–90.
- [5] Sarah TH Low, P Govind Sakhardande, Yi Feng Lai, Andrew DS Long, and Satveer Kaur-Gill. 2021. Attitudes and perceptions toward healthcare technology adoption among older adults in Singapore: a qualitative study. *Frontiers in Public Health* 9 (2021), 588590.
- [6] Carmel Majidi. 2014. Soft robotics: a perspective—current trends and prospects for the future. *Soft robotics* 1, 1 (2014), 5–11.
- [7] Mariangela Manti, Andrea Pratesi, Egidio Falotico, Matteo Cianchetti, and Cecilia Laschi. 2016. Soft assistive robot for personal care of elderly people. In *2016 6th IEEE international conference on biomedical robotics and biomechatronics (BioRob)*. Ieee, 833–838.
- [8] Elaine Mordoch, Angela Osterreicher, Lorna Guse, Kerstin Roger, and Genevieve Thompson. 2013. Use of social commitment robots in the care of elderly people with dementia: A literature review. *Maturitas* 74, 1 (2013), 14–20.
- [9] Tommaso Proietti, Ciaran O'Neill, Lucas Gerez, Tazzy Cole, Sarah Mendelowitz, Kristin Nuckols, Cameron Hohimer, David Lin, Sabrina Paganoni, and Conor Walsh. 2023. Restoring arm function with a soft robotic wearable for individuals with amyotrophic lateral sclerosis. *Science Translational Medicine* 15, 681 (2023), eadd1504.
- [10] Yun Qu. 2020. *Assistive Technologies for Individuals with Alzheimer and Investigating Main Challenges of Adopting the Technologies*. Master's thesis. The University of North Carolina at Charlotte.
- [11] Siri Jakobsson Store, Maria Tillfors, Erik Wästlund, Charlotte Angelhoff, and Annika Norell Clarke. 2022. Technically sleeping? A clinical single-case study of a commercial sleep robot. *Frontiers in Psychology* 13 (2022), 919023.
- [12] I Winblad, P Viramo, A Remes, M Manninen, and J Jokelainen. 2010. Prevalence of dementia—a rising challenge among ageing populations. *European Geriatric Medicine* 1, 6 (2010), 330–333.
- [13] Qiaolian Xie, Qiaoling Meng, Wenwei Yu, Rongna Xu, Zhiyu Wu, Xiaoming Wang, and Hongliu Yu. 2023. Design of a soft bionic elbow exoskeleton based on shape memory alloy spring actuators. *Mechanical Sciences* 14, 1 (2023), 159–170.
- [14] Fitri Yakub, Ahmad Zahran Md Khudzari, and Yasuchika Mori. 2014. Recent trends for practical rehabilitation robotics, current challenges and the future. *International Journal of Rehabilitation Research* 37, 1 (2014), 9–21.